

PROJECT TITLE

Finthropic

The domain of the Project:

Financial Technology and Generative Artificial Intelligence

COURSE NAME

Generative AI

Team Mentors (and their designation):

Mr. Prujith Radhakrishnan — Generative AI Mentor

Team Members:

Mr. Mridul Goyal

Period of the project

May 2026 to July 2026

Organised by SURE Trust

Declaration

The project titled "Finthropic" has been mentored by Mr. Prujith Radhakrishnan, organised by SURE Trust, from May 2026 to July 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the member of the team mentioned below has worked on it successfully and enhanced practical knowledge in the domain.

Team Members:

Mridul Goyal - Mridul Goyal

Mentor's Name: Mr. Prujith Radhakrishnan
Designation: Generative AI Mentor

Prof. Radhakumari
Executive Director & Founder
SURE Trust

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Executive Summary

Finthropic is an AI-powered financial intelligence platform that generates structured equity research reports from publicly available company news. The system ingests financial news through the Finnhub API, scores and ranks articles by relevance and impact, extracts structured market events using Google Gemini, incorporates historical report context for temporal reasoning, and produces narrative research output that is parsed into structured sections for web display.

The implementation consists of a FastAPI backend with PostgreSQL persistence and a Next.js frontend for report generation and viewing. During the project period (May 2026 to July 2026), the core pipeline from news ingestion to structured report delivery was implemented and validated with integration tests and live report generation for companies such as Apple Inc. (AAPL).

The primary outcome is a working end-to-end application that reduces manual effort in synthesizing financial news into analyst-style reports while maintaining evidence grounding through event-level extraction and structured output formatting.

Introduction

Background and Context

Financial analysts and investors routinely process large volumes of news to understand company performance, risks, and opportunities. Manual synthesis of this information is time-consuming and difficult to scale across multiple companies and time periods. Generative AI offers a practical approach to automate structured summarization when combined with reliable data ingestion, ranking, and persistence layers.

Finthropic was developed as a solo project under the Generative AI course, mentored by Mr. Prujith Radhakrishnan, to explore how large language models can be integrated into a production-style backend architecture for financial research automation.

Problem Statement

Existing financial news tools typically present raw articles or single-point summaries without structured analysis across risks, opportunities, and forward-looking outlook. Analysts need reports that organize developments into actionable sections, prioritize high-impact news, and retain context from prior reporting periods.

The problem addressed by Finthropic is: how to automatically transform time-bounded company news into institutional-style research reports with structured sections, grounded event extraction, and reusable historical context.

Scope and Limitations

- Scope covers news ingestion via Finnhub, event extraction, report generation, structured parsing, PostgreSQL storage, and a Next.js UI for report generation and viewing.
- Report types supported: quick, simple, prime, and expert, each with different news volume and historical depth configuration.
- User authentication (JWT-based registration and login) is implemented in the backend but is not integrated into the primary report generation workflow in the current frontend.
- The dashboard and reports listing pages use static demo data and are not yet connected to live backend report retrieval.
- Report generation depends on external API availability (Finnhub, Google Gemini) and configured API keys.
- The system analyzes news summaries from Finnhub rather than full article text for all sources.

Innovation Component

The innovation in Finthropic lies in combining three layers: (1) deterministic news scoring and ranking, (2) LLM-based batch event extraction with structured JSON output, and (3) a historical

context memory layer that stores period summaries and feeds them into subsequent report generation. This pipeline produces repeatable, structured financial intelligence rather than unstructured chat output.

Project Objectives

Primary Objectives

- Design and implement a backend pipeline that ingests company news for a given ticker symbol and date range.
- Extract structured financial events from news using a large language model.
- Generate narrative equity research reports with defined analytical sections.
- Parse generated reports into structured JSON for frontend rendering.
- Persist news, reports, and historical context in a relational database.
- Build a web interface for initiating report generation and viewing results.

Expected Outcomes and Deliverables

- A functional FastAPI backend with report generation, news ingestion, and authentication endpoints.
- A Next.js frontend with landing page, report generation form, and structured report viewer.
- PostgreSQL data model for news, events, reports, historical context, and users.
- Integration and unit tests covering ingestion, event extraction, and report generation.
- Documented system architecture and working demonstration with live report output.

Methodology and Results

Methods and Technology Used

The project follows a layered architecture with separate presentation, API, business logic, and data tiers. The backend uses service-oriented modules for ingestion, scoring, event extraction, context building, report generation, parsing, and persistence. The frontend communicates with the backend through a REST API using Axios and TanStack React Query.

Layer	Technology	Role
Frontend	Next.js 16, React 19, TypeScript	UI and report display
Styling	Tailwind CSS 4	Responsive dark-mode interface
Backend	FastAPI, Uvicorn	REST API server
Database	PostgreSQL, SQLAlchemy 2	Persistent storage
LLM	Google Gemini (google-genai)	Event extraction and report generation
News Data	Finnhub API	Company news ingestion
Auth	JWT, bcrypt, python-jose	User registration and login
Testing	pytest	Unit and integration tests

Tools and Software Used

- Python 3.13 with FastAPI 0.139, Pydantic 2, SQLAlchemy 2, requests, python-dotenv.
- Node.js ecosystem with Next.js 16, React Query, Axios, react-markdown.
- PostgreSQL database (local instance on port 5433).
- Google Gemini API for LLM inference.
- Finnhub API for financial news retrieval.
- pytest for automated testing.
- Brave browser for application testing and screenshot capture.

Data Collection Approach

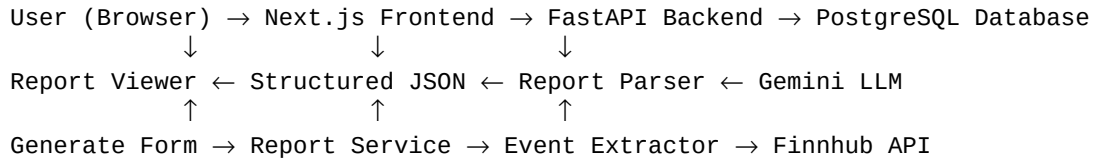
Financial news is collected programmatically through the Finnhub company-news endpoint. When a report is requested, the IngestionService fetches articles for the specified ticker symbol and date range. Each article is deduplicated by finnhub_id before insertion into the news table. Stored fields include headline, summary, source, URL, and published timestamp.

After ingestion, the NewsScoringService ranks articles using a weighted composite score based on relevance (35%), recency (25%), source credibility (15%), and financial keyword impact (25%). The top-k articles are selected according to report type configuration: quick (10), simple

(20), prime (50), and expert (100).

Project Architecture

High-Level System Architecture



The report generation workflow executes the following steps: (1) check for an existing cached report matching symbol, date range, and report type; (2) ingest news from Finnhub; (3) query stored news for the date window; (4) score and select top articles; (5) batch-extract events via Gemini; (6) build event context text; (7) retrieve historical context summaries from prior periods; (8) generate the narrative report via Gemini; (9) parse the report into structured JSON; (10) generate and store a historical summary; (11) persist the report and return the response.

Database Tables

Table	Purpose
news	Stores ingested Finnhub articles with deduplication by finnhub_id
events	Stores extracted event metadata linked to news records
reports	Stores generated report text and structured JSON output
historical_context	Stores period summaries for temporal context
users	Stores registered user accounts with hashed passwords

Primary API Endpoints

Endpoint	Method	Description
/reports/generate	POST	Generate or retrieve a research report
/news/ingest	POST	Ingest news for a symbol and date range
/auth/register	POST	Register a new user
/auth/login	POST	Authenticate and receive JWT token
/auth/me	GET	Return current authenticated user
/	GET	Health check endpoint

Application Screenshots and Results

The following screenshots demonstrate the working Finthropic application. A sample report was generated for Apple Inc. (AAPL) for the period 1 June 2026 to 30 June 2026 using the Prime report type.

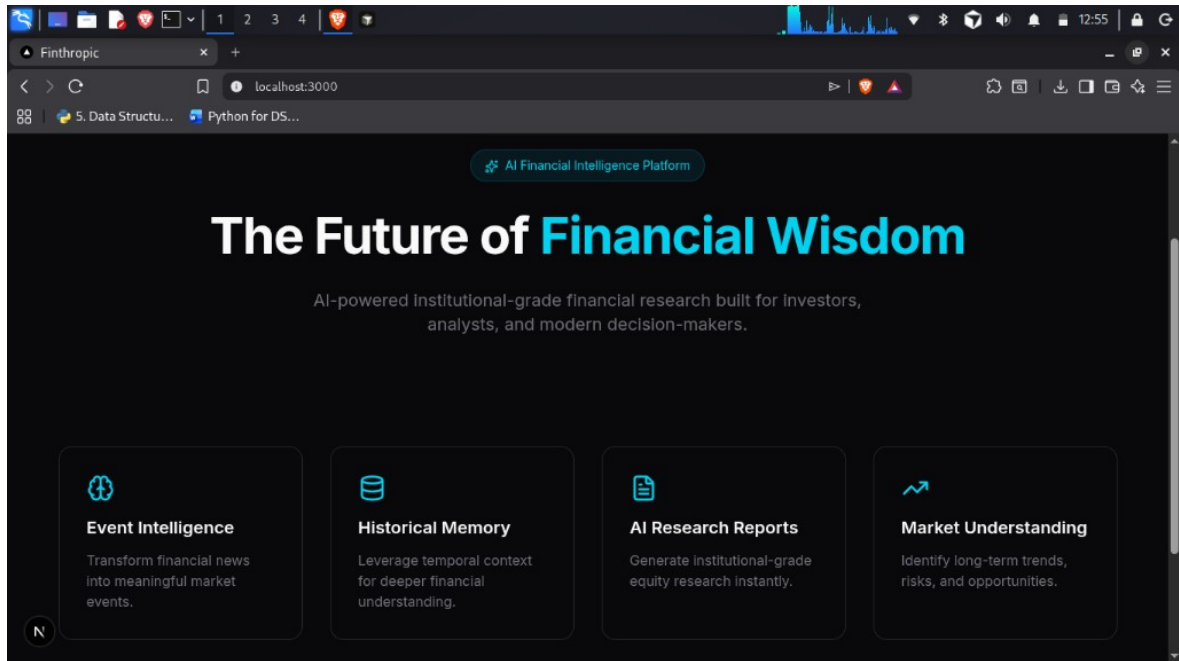


Figure 1: Finthropic landing page showing the platform overview and core capabilities.

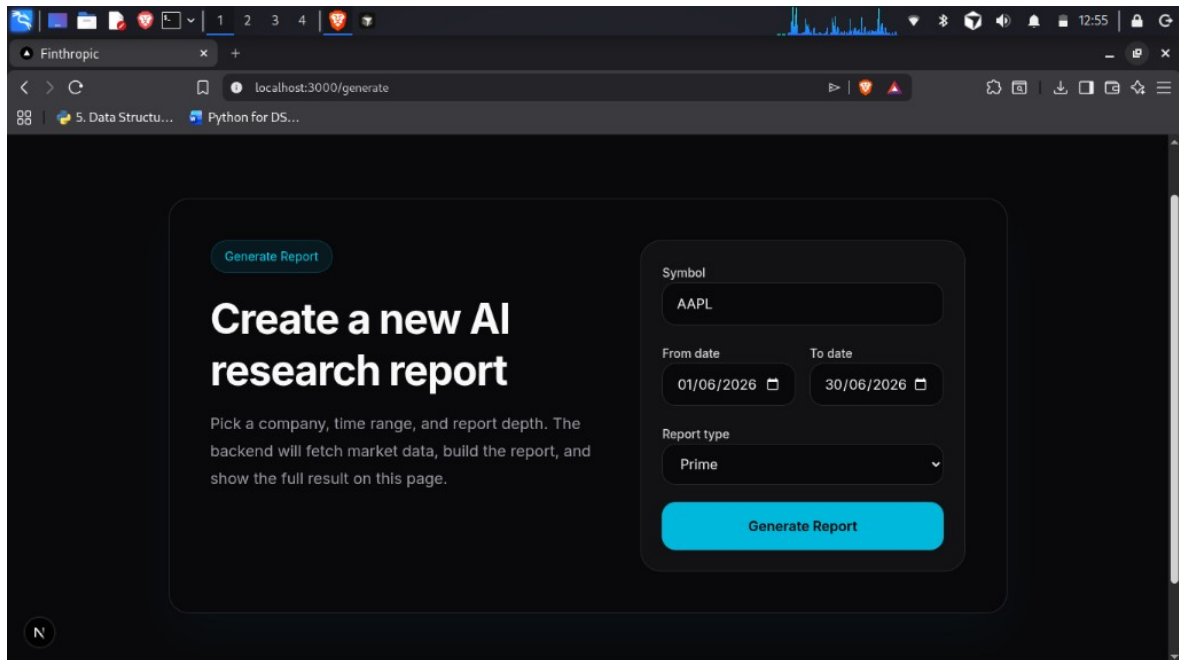


Figure 2: Report generation interface where users enter symbol, date range, and report type.

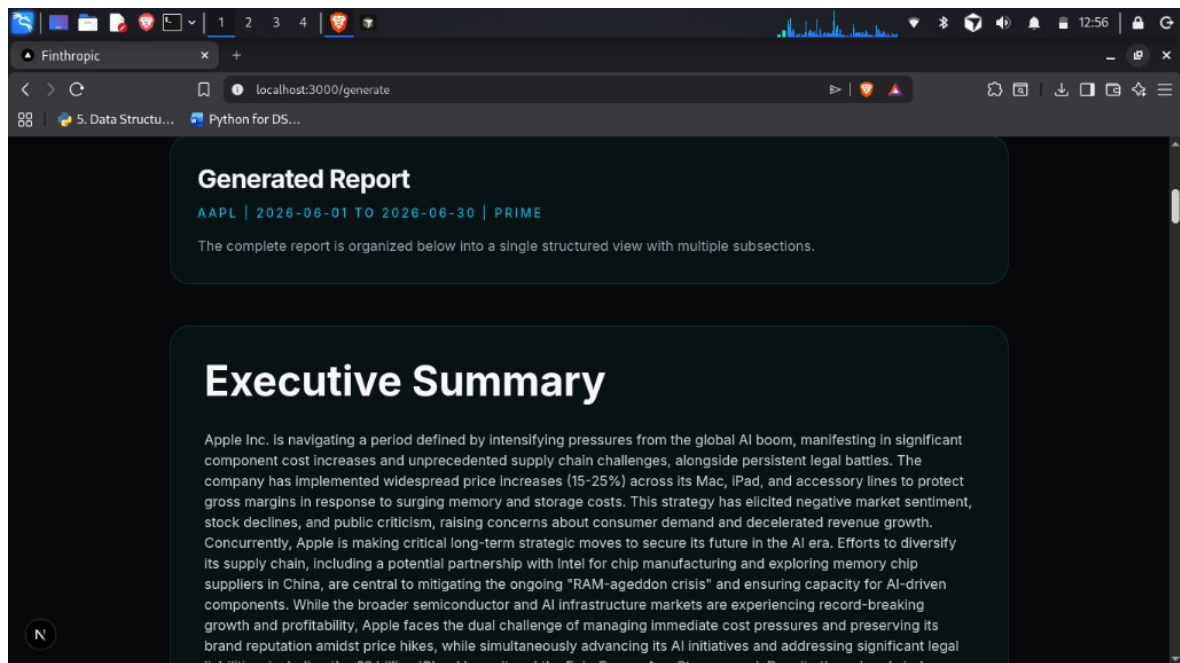


Figure 3: Generated report header with executive summary for AAPL (Prime, June 2026).

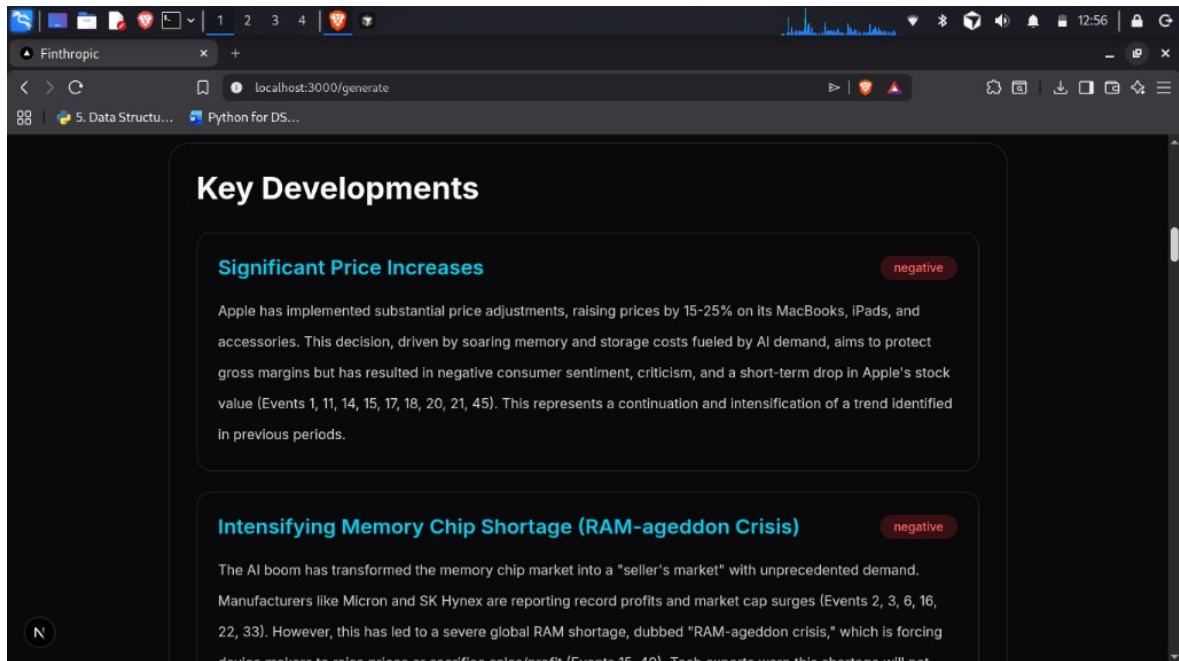


Figure 4: Structured Key Developments section with event-referenced analysis.

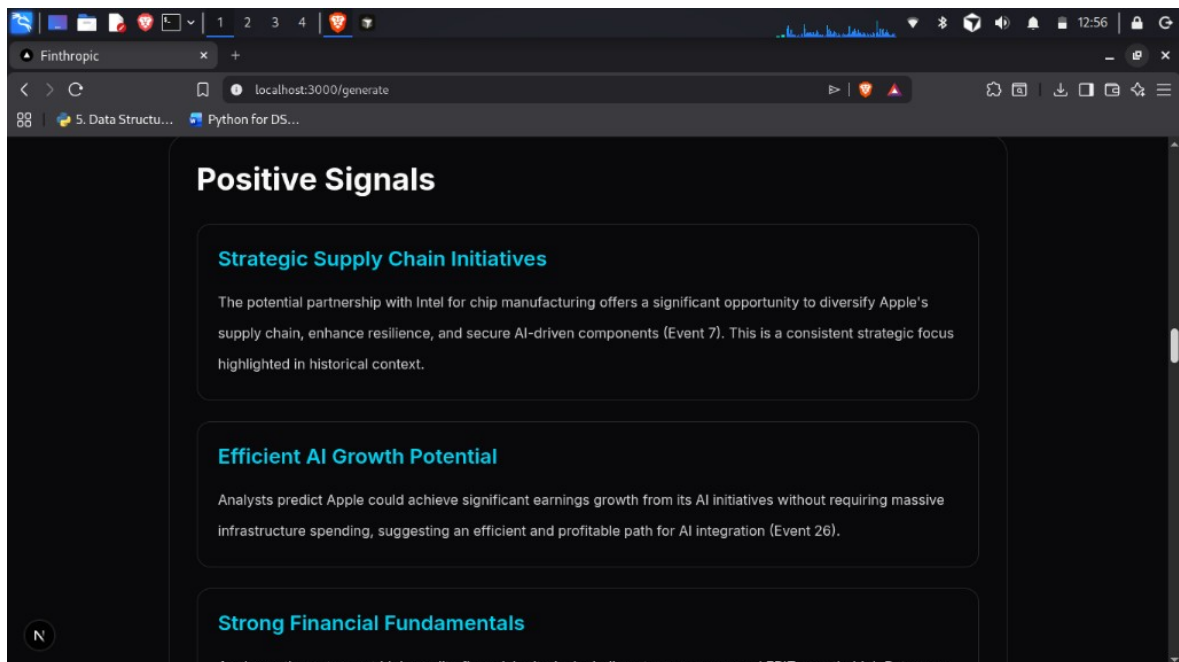


Figure 5: Positive Signals section identifying favorable developments.

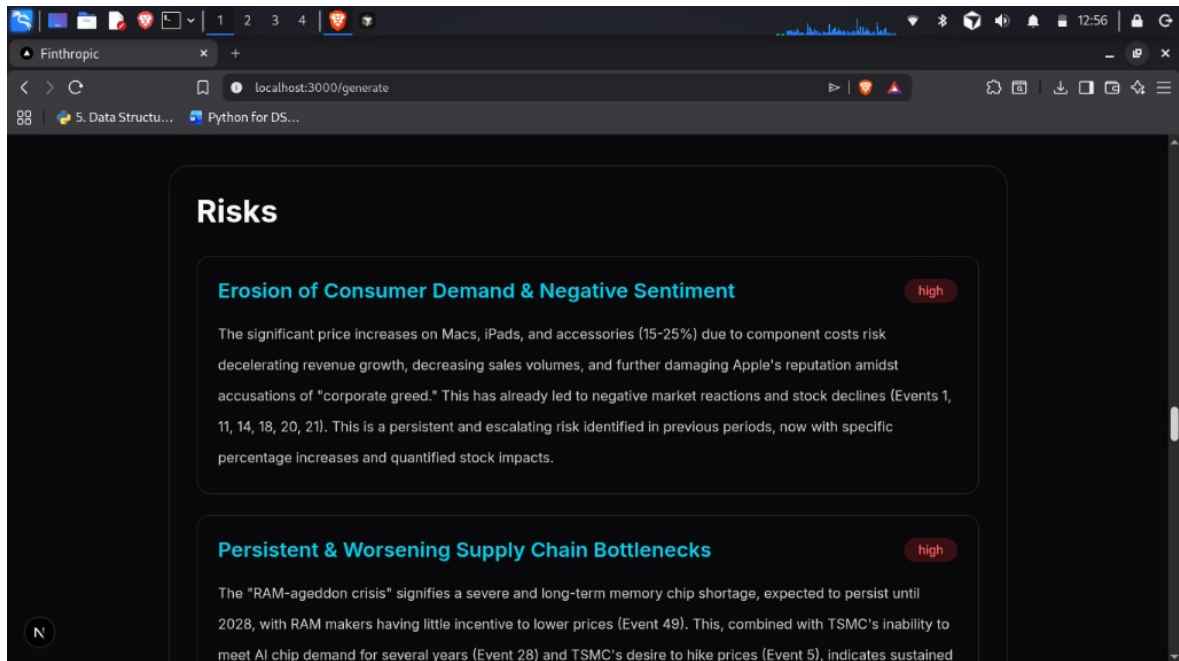


Figure 6: Risks section with severity indicators for identified concerns.

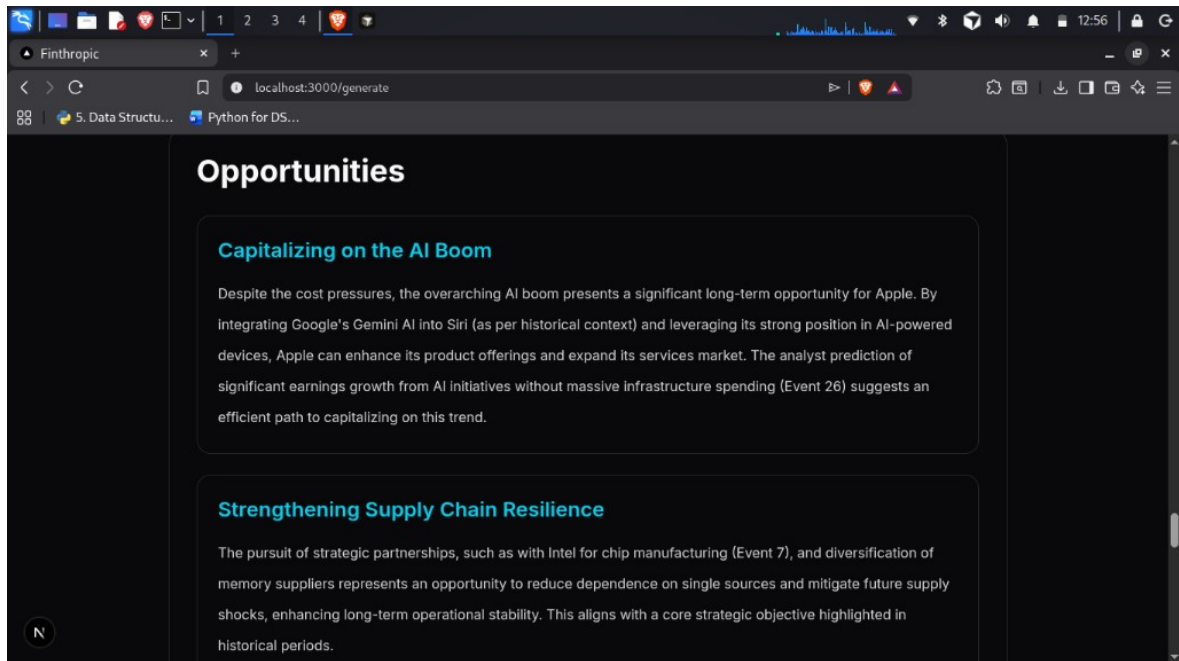


Figure 7: Opportunities section highlighting strategic growth areas.

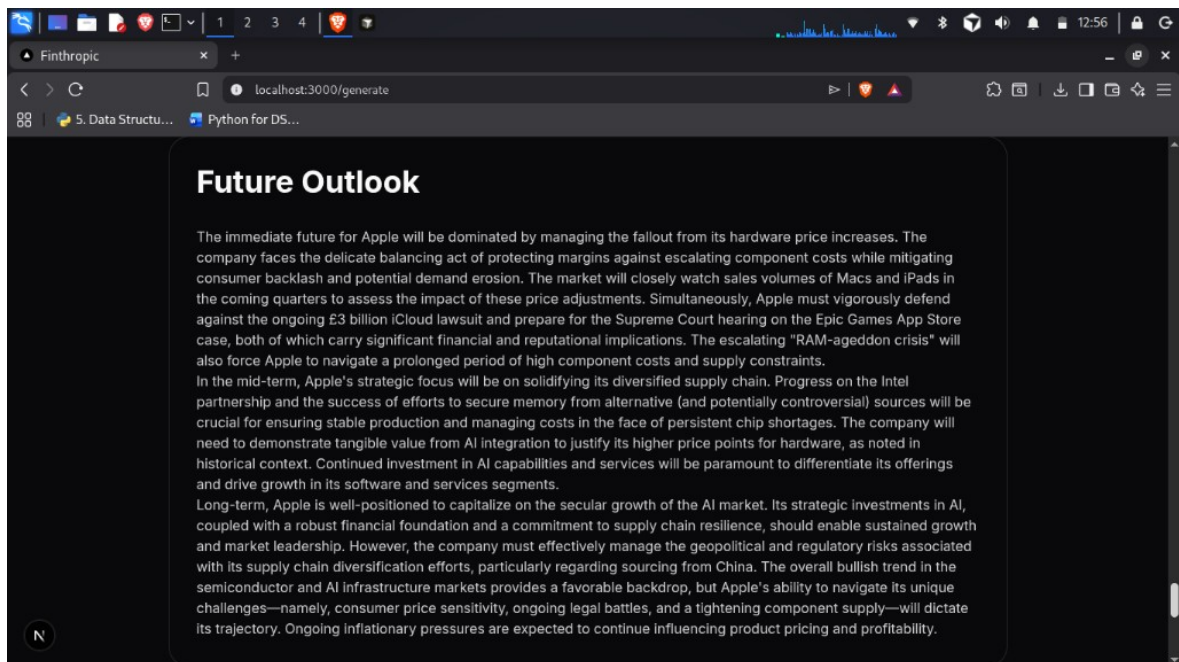


Figure 8: Future Outlook section with forward-looking assessment.

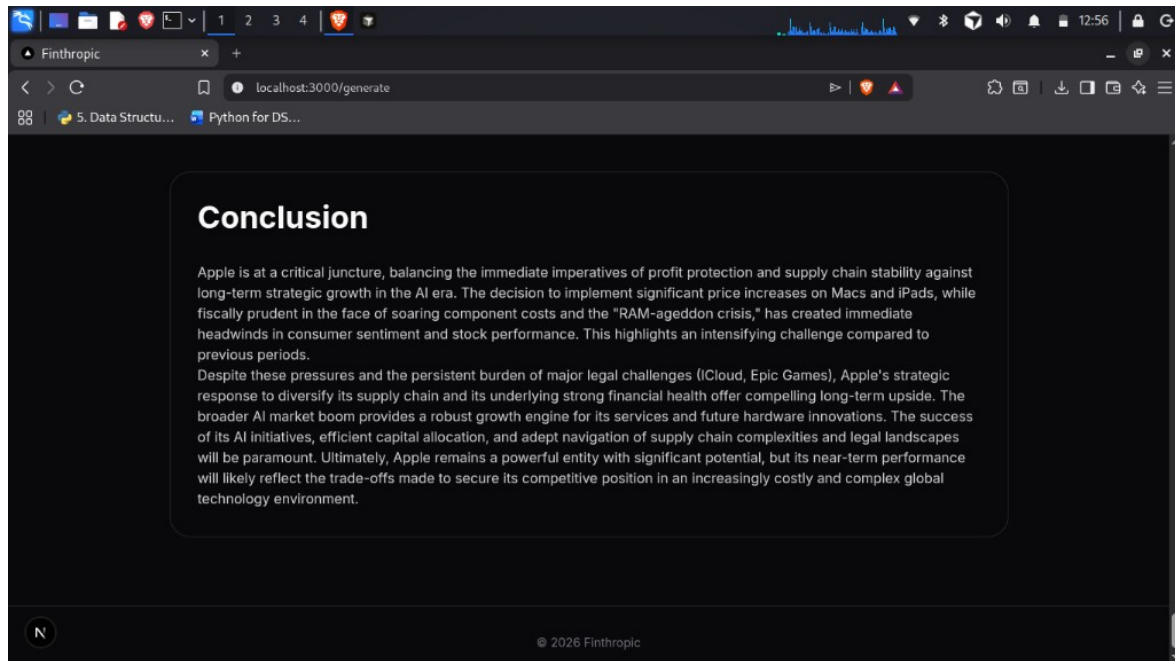


Figure 9: Conclusion section providing balanced final assessment.

Project GitHub Link

Project	GitHub	Link:
https://github.com/sure-trust/MRIDULTECH-g4-gen-ai/tree/main/Final%20capstone%20project		

Social / Industry Relevance of the Project

Finthropic addresses a practical need in the financial services and investment research domain: reducing the manual effort required to monitor company news and produce structured analysis. For retail investors, independent analysts, and early-career finance professionals, automated research synthesis can improve access to organized market intelligence.

The project demonstrates how generative AI can be applied responsibly in a business context when combined with structured data pipelines, source attribution through event extraction, and explicit report sectioning. Rather than replacing human judgment, the system supports faster initial research and consistent report formatting.

From an industry perspective, the architecture patterns used—external API integration, LLM orchestration, structured output parsing, and historical context retention—are applicable beyond finance to any domain requiring time-aware document synthesis.

Learning and Reflection

Mr. Mridul Goyal — Technical Learnings

- Backend development with FastAPI, including router organization, service-layer architecture, and Pydantic schema validation.
- Managing AI pipelines spanning data ingestion, LLM prompting, JSON parsing, error handling, and retry logic for API reliability.
- LLM API integration and usage with Google Gemini for batch event extraction, report generation, and structured output parsing.
- Leveraging generative AI to build scalable solutions that reduce manual effort in conventional research workflows.
- Basic system design practices including layered architecture, database modeling, and separation of concerns across modules.
- Backend security fundamentals including password hashing with bcrypt, JWT token generation, and protected route implementation.

Mr. Mridul Goyal — Overall Experience

This solo project provided end-to-end experience in building a full-stack AI application from problem definition through working demonstration. The most valuable learning was understanding how to structure an LLM-powered pipeline so that model output remains grounded in ingested data rather than producing unconstrained text. Integrating multiple services—news APIs, database persistence, LLM calls, and a modern frontend—required careful sequencing and iterative debugging. The project reinforced that effective generative AI applications depend as much on data engineering and system design as on prompt quality.

Conclusion and Future Scope

Objectives and Achievements

Finthropic successfully achieved its core objective of transforming company news into structured AI-generated research reports. The implemented pipeline covers news ingestion, relevance scoring, event extraction, historical context integration, report generation, structured parsing, and web-based presentation. Live testing with AAPL demonstrated that the system produces coherent, multi-section reports with event-referenced analysis.

Future Scope

- Connect the dashboard and reports listing pages to live backend data instead of static demo content.
- Integrate JWT authentication into the frontend report workflow.
- Add a report listing and retrieval API endpoint for stored reports.
- Persist extracted events to the events table during the pipeline.
- Support additional news sources beyond Finnhub.
- Add export functionality (PDF/Markdown download) for generated reports.
- Implement rate limiting, caching, and production deployment configuration.
- Expand test coverage and add monitoring for LLM API failures.

Finthropic establishes a functional foundation for AI-assisted financial research. With continued development, it can evolve into a more complete research platform while maintaining the evidence-grounded pipeline architecture implemented during this project.